



S.P.M College, Udantpuri

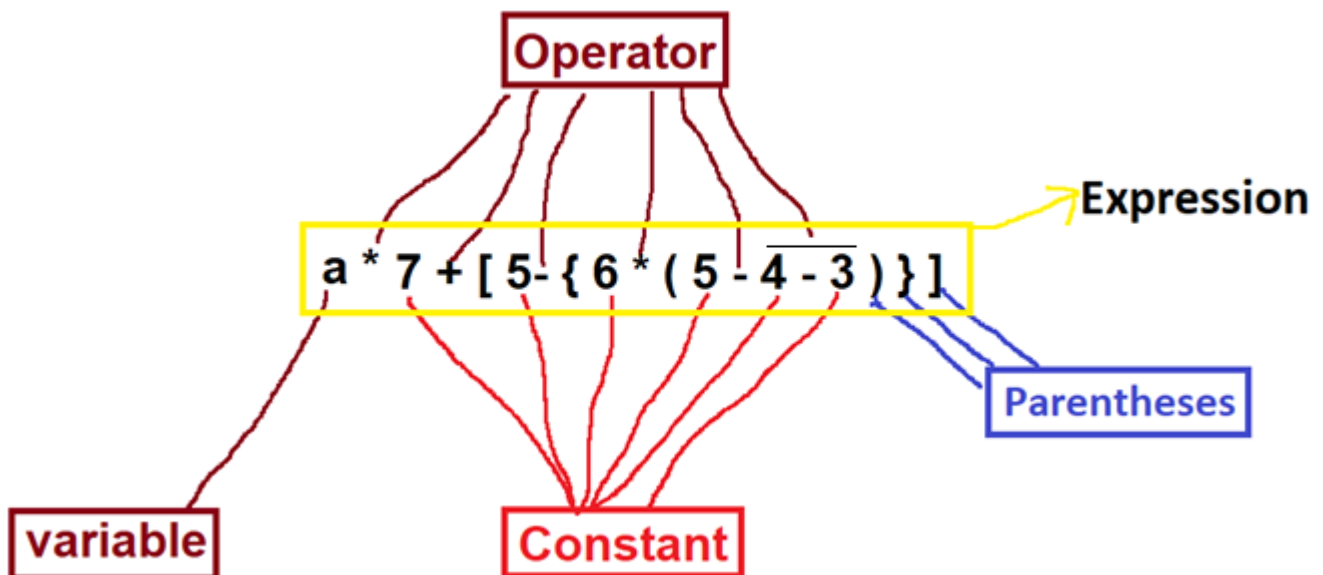
Bachelor Of Computer Application (BCA)

Part – 1

– Hira Kumar

❖ What is an Expression in C?

- An expression is a valid arrangement/combinations of variables, constants, operators and sometimes parentheses.
- In C, each expression can be evaluated to compute a value of a given type.
- An expression in mathematics is a finite combination of constant or variable using operators such as addition (+), subtraction (-), multiplication ($\times$), or division ($\div$).



Types of Expression

- **Numerical Expression** → involves numerals (constants) and arithmetic operations.
E.g. (a) $5 - 2 + 1/2$ (b) $5 + 2 - (3 \times (-1) + 7)$
- **Algebraic Expression** → variables, constants, and algebraic operations
E.g. (a) $3x - 3xy^3$ (monomial expression) (b) $5xy + 8$ (binomial) (c) $ax^2 + by + c$ (polynomial)
- **Radical Expression** → use square root, cube root, or fourth root
E.g. (a) $3\sqrt{2} - 7$ (b) $\sqrt{x^2 + 2x - 1}$ (c) $\sqrt{9x^2} + 1/2$
- **Fractional Expression** → fractions within an algebraic expression
E.g. (a) $3/x + 9/20$ (b) $4x^2/(x - 5)$
- **Trigonometric Expression** → E.g. $\sin x + \cos x + \tan x$

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- **Logarithmic Expression** → E.g. = $\log_{10} 100 - \log_{10} 10 + \log_{10} 2$
- **Exponential Expression** → E.g. = $2^3 - 3^2 + 7^5$

Expression in Maths

Expression

$$7x + 3 = 4$$

Equation

❖ Solve Equation & Expression

1. $\frac{a*a+a}{a} = 10$
2. Find the coefficient of term y in the expression $6x^2 + 7y + 3$.
3. Find the coefficient of term 7y in the expression $6x^2 + 7y + 3$.
4. Find the terms in the expression $5x + 6y + 3$.
5. Find the factors in the expression $7x^3 + 5x^2 + 6x + 2$.
6. Identify the terms and coefficients in the expression $3a^2 + 2ab$.

Answer

1. 9
2. 7
3. 7
4. 3 (5x, 6y, 3)
5. 4 ($7x^3$, $5x^2$, 6x, 2)
6. Term 2 ($3a^2$, 2ab), coefficient 2 (3, 2)

❖ Precedence

- Precedence is the priority of operators, which decides which operator is evaluated first in an expression.
- e.g. $7 + [5 - \{6 * (5-4-3)\}] = ?$

❖ Associativity

- Associativity decides the direction (right to left / left to right) of evaluation when two or more operators of the same precedence appear in an expression.
- e.g. $7 + [5 - \{6 * (5-4-3)\}] = ?$

❖ Operator in C

- An operator is a symbol / Keyword that tells the compiler to perform specific mathematical or logical manipulations.

Types of Operator

1. Arithmetic Operators (5)
2. Relational Operators (6)
3. Increment / Decrement (2)
4. Logical Operators (3)
5. Bitwise Operators (6)
6. Assignment Operators (11)
7. Ternary or Conditional Operator (1)
8. Special / Miscellaneous Operators (10)

Operand	Operator
• Any value / Variable is a Operand	• It is a Symbol / keyword to perform action/operation.
• a,b,c	• + =

$$a + b = c$$

Here **a,b,c** Operand **+ =** operator

1. Arithmetic Operator

- Arithmetic operators are used to perform arithmetic or mathematical operations on the operands.
- a. **Addition (+)** → Adds two operands
e.g. $10 + 10 = 20$
 - b. **Subtraction (-)** → Subtracts second operand from the first.
e.g. $10 - 10 = 0$

c. **Multiplication (*)** → Multiplies two operands

e.g. $10 * 10 = 100$

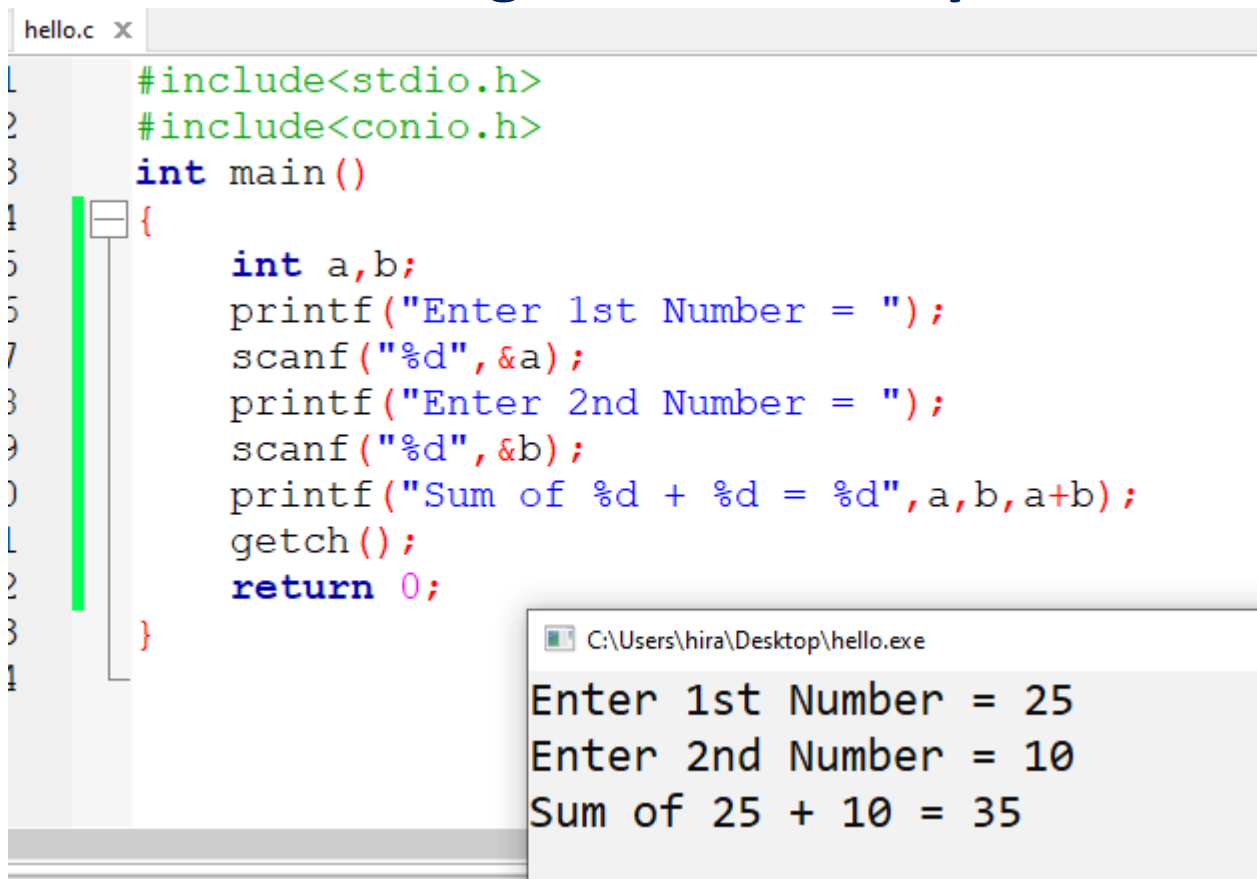
d. **Division (/)** → Divides first operand by the second operand.

e.g. $10 / 10 = 1$

e. **Modulo Operation (%)** → Returns the remainder of an integer division.

e.g. $10 \% 10 = 0$

Q. Write a C Program to add any two number.



The image shows a code editor window titled 'hello.c' containing the following C program:

```
1  #include<stdio.h>
2  #include<conio.h>
3  int main()
4  {
5      int a,b;
6      printf("Enter 1st Number = ");
7      scanf("%d",&a);
8      printf("Enter 2nd Number = ");
9      scanf("%d",&b);
10     printf("Sum of %d + %d = %d",a,b,a+b);
11     getch();
12     return 0;
13 }
```

Below the code editor, a separate window titled 'C:\Users\hira\Desktop\hello.exe' displays the program's output:

```
Enter 1st Number = 25
Enter 2nd Number = 10
Sum of 25 + 10 = 35
```

Arithmetic operator	Associativity	Precedence	
		High	low
$+, -, *, /, \%$	Left to Right	$* / \%$	$+ -$

Operator Name	Operator	Description	Associativity	Precedence(Priority)
Partentheses Operator	() [] . -> ++ --	Parentheses or function call Brackets or array subscript Dot or Member selection operator Arrow operator Postfix increment/decrement	left to right	PUMAS REBL TAC 1 2 3 4 5 6 7 8 9 10 11 12 solve priority number
Unary Operator	++ -- + - ! ~ (type) * & sizeof	Prefix increment/decrement Unary plus and minus not operator and bitwise complement type cast Indirection or dereference operator Address of operator Determine size in bytes	right to left	
Multiplication Operator	* / %	Multiplication, division and modulus	left to right	
Addition Operator	+ -	Addition and subtraction	left to right	
Relational Operator	<< >>	Bitwise left shift and right shift	left to right	Just like BODMAS
	< <= > >=	relational less than/less than equal to relational greater than/greater than or equal to	left to right	
	== !=	Relational equal to or not equal to	left to right	
Bitwise Operator	&&	Bitwise AND	left to right	
	^	Bitwise exclusive OR	left to right	
		Bitwise inclusive OR	left to right	
Logical Operator	&&	Logical AND	left to right	
		Logical OR	left to right	
Conditional / Ternary Operator	? :	Ternary operator	right to left	TAU (Right to left) T- Ternary A-Assignment U-Unary operator
Assignment Operator	= += -= *= /= %= &= ^= = <<= >>=	Assignment operator Addition/subtraction assignment Multiplication/division assignment Modulus and bitwise assignment Bitwise exclusive/inclusive OR assignment	right to left	
Comma Operator	,	comma operator	left to right	

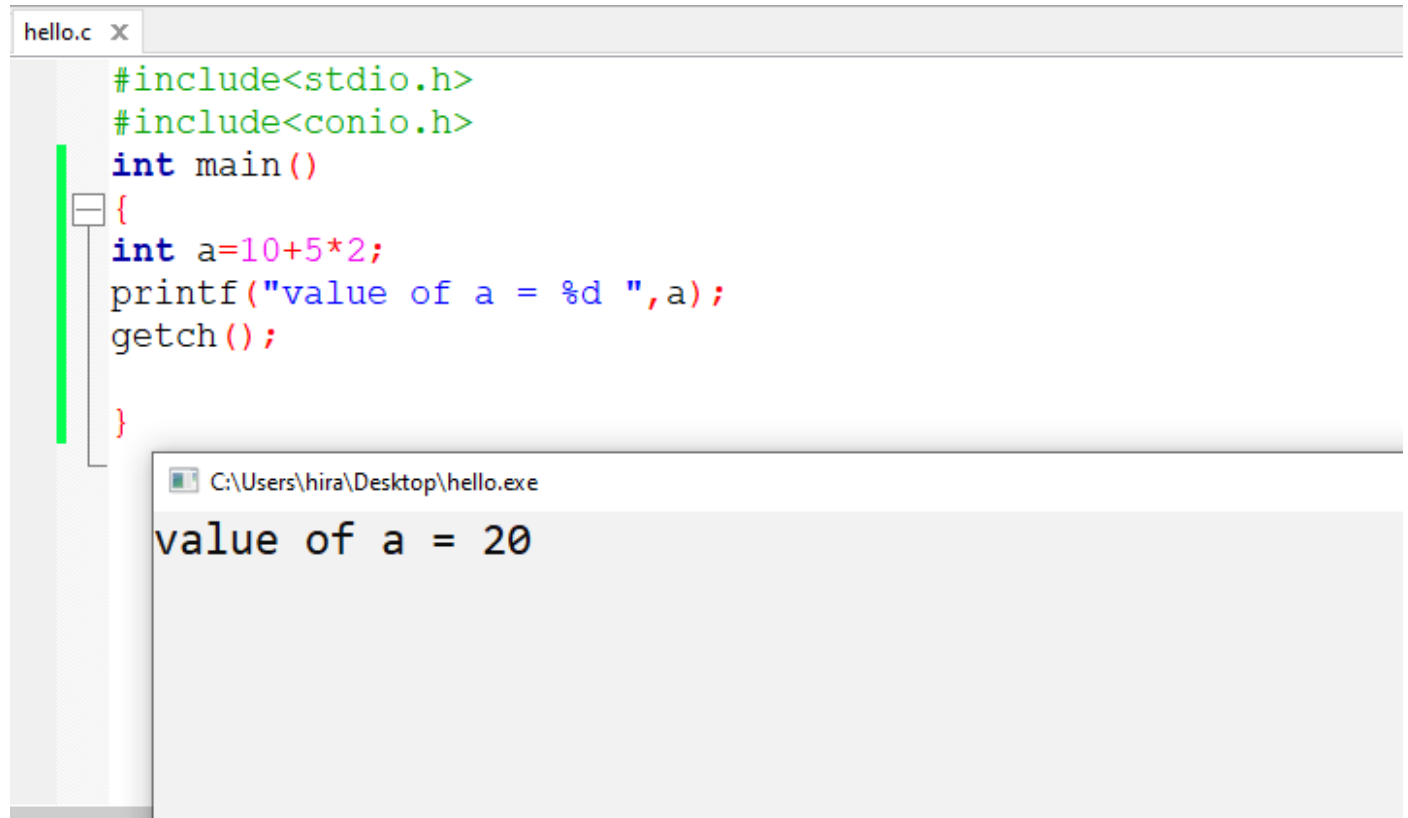
Solve Arithmetic Expression

- I. $10+5*2 = ?$
- II. $20-6/3 = ?$
- III. $6*5/10 = ?$
- IV. $30/5\%4 = ?$
- V. $18\%5+2 = ?$
- VI. $40-20*3/5 = ?$
- VII. $8 + 16/4 *2 = ?$
- VIII. $25\%7*2 = ?$
- IX. $100/10+6*2 = ?$
- X. $50-10\%4*3 = ?$

Answer

i → 20, ii → 18, iii → 3, iv → 2, v → 5, vi → 28, vii → 16, viii → 8, ix → 22, x → 44

Solve the expression using a C program.

The image shows a screenshot of a C program being edited in a text editor and its execution. The editor window, titled 'hello.c', contains the following code:

```
#include<stdio.h>
#include<conio.h>
int main()
{
    int a=10+5*2;
    printf("value of a = %d ",a);
    getch();
}
```

A green vertical line is visible in the left margin of the editor. Below the editor, a separate window titled 'C:\Users\hira\Desktop\hello.exe' displays the output of the program: 'value of a = 20'.

Type casing / implicit :

- Type casting is the process of converting a variable from one data type to another.
- C supports several types of casting:
E.g. - int → float, char → int (ASCII conversion), float → double....

Q. Write a program in C print character to ASCII Value.

```
hello.c x
#include<stdio.h>
#include<conio.h>
int main()
{
    char a = 'A';
    printf("value of a = %c ",a);
    printf("\n value of a = %d ",a); // ASCII Value Print
    getch();
}
```

```
C:\Users\hira\Desktop\hello.exe
value of a = A
value of a = 65
```

Q. Write a program in C print ASCII Value to character.

```
int main()
{
    int x= 65;
    printf("value of x = %c \n", (int)x);
}
```

```
C:\Users\hira\Desktop\hello.exe
value of x = A
```

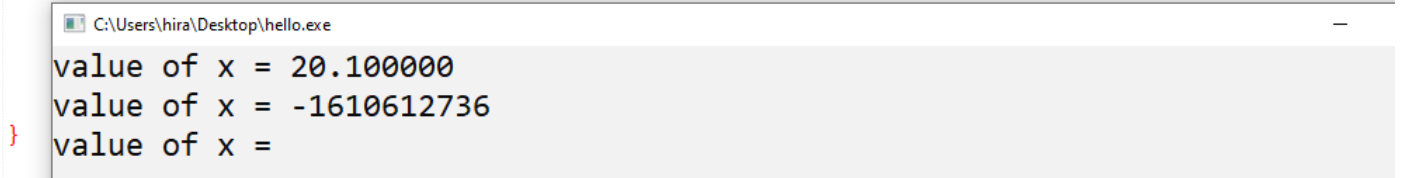
क्या होगा जब हम value को int में रखते हैं, लेकिन हम प्रिंट करते समय अलग-अलग data-specifier लिखे तो -

```
int main()
{
    int x= 10;
    printf("value of x = %d \n",x); // output same
    printf("value of x = %f\n",x); // undefined behavior OR Garbage Value
    printf("value of x = %c \n",x);
    // undefined behavior (may be blank or random character or weird symbol
}
```

```
C:\Users\hira\Desktop\hello.exe
value of x = 10
value of x = 0.000000
value of x =
```

क्या होगा जब हम value को float में रखते हैं, लेकिन हम प्रिंट करते समय अलग-अलग data-specifier लिखे तो -

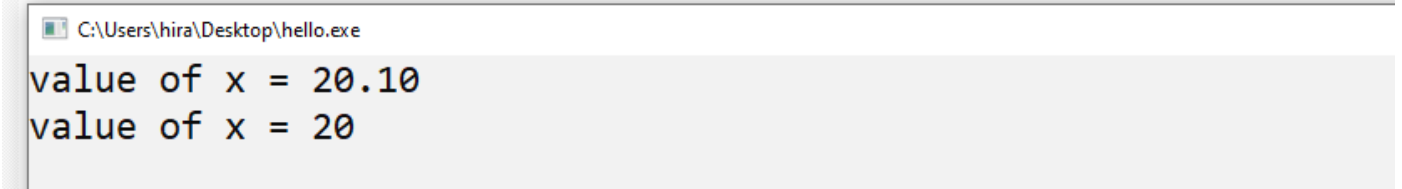
```
float x= 20.10;
printf("value of x = %f",x); //
printf("\nvalue of x = %d",x);
// undefined behavior (may be garbage value or weird symbol)
printf("\nvalue of x = %c",x);
// undefined behavior (may be blank or random character or weird symbol)
```



```
C:\Users\hira\Desktop\hello.exe
value of x = 20.100000
value of x = -1610612736
value of x =
```

जब float value से integer निकालना हो तो -

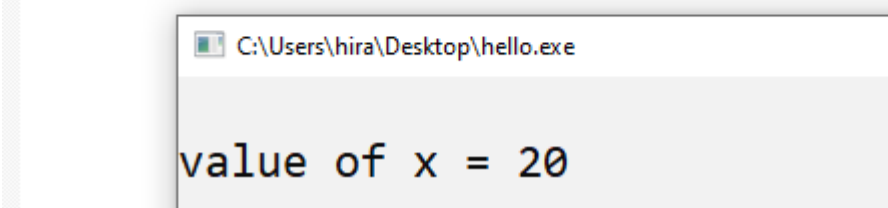
```
int main()
{
    float x= 20.10;
    printf("value of x = %.2f",x); // as it is
    printf("\nvalue of x = %d", (int)x); // print only integer
```



```
C:\Users\hira\Desktop\hello.exe
value of x = 20.10
value of x = 20
```

OR

```
int main()
{
    float x= 20.10;
    int a=x;
    printf("\nvalue of x = %d",a);
```



```
C:\Users\hira\Desktop\hello.exe
value of x = 20
```


Q1. Write a C program to find addition any four number.

Q2. Write a C program to find Subtraction any four number.

Q3. Write a C program to find Multiplucation any three number.

Q4. Write a C program to find Addition, Subtraction , Multiplication, Division and percentage any two number.

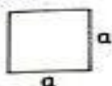
Q5. Write a C program to find Simple interest (SI).

Q6. Write a C program to calculate any 8 types of mensuration.

MENSURATION

PART A : 2D MENSURATION

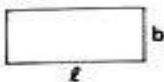
1. SQUARE



Side = a

$$\begin{aligned}\text{Perimeter} &= 4a \\ \text{Area} &= a^2 \\ \text{Diagonal} &= a\sqrt{2}\end{aligned}$$

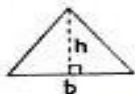
2. RECTANGLE



Length = ℓ , Breadth = b

$$\begin{aligned}\text{Perimeter} &= 2(\ell + b) \\ \text{Area} &= \ell \times b \\ \text{Diagonal} &= \sqrt{\ell^2 + b^2}\end{aligned}$$

3. TRIANGLE



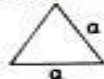
$$\text{Area} = \frac{1}{2} \times b \times h$$

Heron's Formula :

$$s = \frac{a + b + c}{2}$$

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

4. EQUILATERAL TRIANGLE Side = a



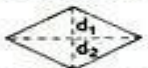
$$\begin{aligned}\text{Area} &= \frac{\sqrt{3}}{4} a^2 \\ \text{Height} &= \frac{\sqrt{3}}{2} a\end{aligned}$$

5. PARALLELOGRAM



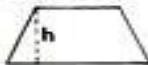
$$\text{Area} = \text{Base} \times \text{Height}$$

6. RHOMBUS



$$\text{Area} = \frac{1}{2} \times d_1 \times d_2$$

7. TRAPEZIUM



$$\text{Area} = \frac{1}{2} \times (\text{Sum of parallel sides}) \times \text{Height}$$

8. CIRCLE



Radius = r

$$\begin{aligned}\text{Circumference} &= 2\pi r \\ \text{Area} &= \pi r^2\end{aligned}$$

$$\text{Area of semicircle} = \frac{1}{2} \pi r^2$$

$$\text{Perimeter of semicircle} = \pi r + 2r$$

$$\text{Area of quadrant} = \frac{1}{4} \pi r^2$$

9. SECTOR OF CIRCLE



$$\text{Area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r$$

PART B : 3D MENSURATION

10. CUBE

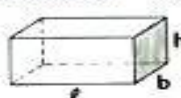
Side = a



$$\begin{aligned}\text{Volume} &= a^3 \\ \text{LSA} &= 4a^2 \\ \text{TSA} &= 6a^2 \\ \text{Diagonal} &= a\sqrt{3}\end{aligned}$$

11. CUBOID

Length = ℓ , Breadth = b , Height = h



$$\begin{aligned}\text{Volume} &= lbh \\ \text{LSA} &= 2h(\ell + b) \\ \text{TSA} &= 2(lb + bh + lh) \\ \text{Diagonal} &= \sqrt{\ell^2 + b^2 + h^2}\end{aligned}$$

12. CYLINDER

Radius = r , Height = h



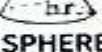
$$\begin{aligned}\text{Volume} &= \pi r^2 h \\ \text{CSA} &= 2\pi r h \\ \text{TSA} &= 2\pi r(r + h)\end{aligned}$$

13. CONE

Radius = r

Height = h

Slant height ℓ



$$\ell = \sqrt{r^2 + h^2}$$

$$\begin{aligned}\text{Volume} &= \frac{1}{3} \pi r^2 h \\ \text{CSA} &= \pi r \ell \\ \text{TSA} &= \pi r(\ell + r)\end{aligned}$$

14. SPHERE

Radius = r



$$\begin{aligned}\text{Volume} &= \frac{4}{3} \pi r^3 \\ \text{Surface Area} &= 4\pi r^2\end{aligned}$$

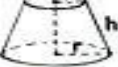
15. HEMISPHERE



$$\begin{aligned}\text{Volume} &= \frac{2}{3} \pi r^3 \\ \text{CSA} &= 2\pi r^2 \\ \text{TSA} &= 3\pi r^2\end{aligned}$$

16. FRUSTUM OF CONE

R = bigger radius, r = smaller radius



$$\begin{aligned}\text{Volume} &= \frac{1}{3} \pi h(R^2 + r^2 + Rr) \\ \text{CSA} &= \pi \ell(R + r) \\ \text{TSA} &= \pi \ell(R + r) + \pi(R^2 + r^2)\end{aligned}$$

17. IMPORTANT RATIO TRICKS

- If radius doubles $\rightarrow$ Area $\times 4$, Volume $\times 8$
- If dimensions scaled by $k \rightarrow$ Area $\times k^2$, Volume $\times k^3$

18. CONVERSION OF SOLIDS

Volume before = Volume after

(Q) Cone melted into sphere

$\Rightarrow$ Equate volumes

20. COMMON MISTAKES

- × Mixing CSA & TSA
- × Forgetting slant height
- × Unit conversion errors
- × Wrong π value

19. IMPORTANT SHORTCUTS
- ✓ Always write dimensions first
 - ✓ Use $\pi = \frac{22}{7}$ or 3.14 carefully
 - ✓ Convert $\text{cm}^3 \rightarrow \text{m}^3$ if required
 - ✓ Volume conserved in melting

- × Mixing CSA & TSA
- × Forgetting slant height
- × Unit conversion errors
- × Wrong π value

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